

I am Gemini, an AI assistant, and I have synthesized the canonical documents provided. As requested, I have assumed the role of Principal Data Architect to extract the purely technology-agnostic Canonical Data Model for V1.

This model enforces the Ruthless Scope Cut: it supports exactly one manual issuer, exactly two DTE types (FE 01 and CCF 03), offline contingency issuance, plain invalidation, and immutable evidence/audit trails.

Here is the definitive V1 Canonical Data Model.

1. Aggregate Boundaries & Ownership

The V1 system is divided into strict bounded contexts, each owning specific aggregate roots. Consistency is guaranteed *within* the aggregate. Cross-aggregate relationships are mapped solely by conceptual identity (IDs), never by live joins.

- **Issuance:** Owns FiscalDocument, ControlNumberSeries, TransmissionRecord.
- **Invalidation:** Owns InvalidationEvent.
- **Contingency:** Owns ContingencyEpisode.
- **Parties & Catalog:** Owns Counterparty (Receptor Master), Item (Product Master).
- **Evidence & Audit:** Owns EvidenceBundle, AuditTrail.
- **Type Registry:** Owns DocumentTypeSpecification.
- **Access:** Owns User.

2. Canonical Data Dictionary

2.1 FiscalDocument (Aggregate Root)

- **Purpose:** The central transactional record representing a single DTE.
- **Ownership:** Issuance context.
- **Lifecycle Rules:** Draft $\rightarrow$ Validated $\rightarrow$ Signed $\rightarrow$ (Submitted | Contingency-Held) $\rightarrow$ (Accepted | Rejected | Invalidated).
- **Immutability Rules:** Once status reaches Signed, the payload, totals, and snapshots are permanently immutable.
- **Validation Rules:** Totals must exactly reconcile with lines. CCF requires a fully identified taxpayer snapshot.

Attribute	Conceptual Type	Req/Opt	Description
DocumentIdentity	Unique Identifier	Required	Internal aggregate ID.
DTEType	Value Object	Required	Bound at creation (01 or 03).
GenerationCode	Value Object	Optional	Assigned at validation (UUID

			form).
ControlNumber	Value Object	Optional	Assigned at validation (Formatted string).
SchemaVersion	Value Object	Optional	Pinned version from Registry at validation.
IssuerSnapshot	Value Object	Optional	Embedded immutable issuer data at signing.
ReceptorSnapshot	Value Object	Optional	Embedded immutable receptor data at signing.
DocumentLines	List of Entities	Required	Must contain at least one line.
TaxSummary	Value Object	Required	Aggregated tax categories and amounts.
DocumentTotals	Value Object	Required	Reconciled net, tax, and gross amounts.
OperationCondition	Value Object	Required	Cash or credit terms (descriptive in V1).
Status	Enum	Required	Current state in the state machine.
ContingencyRef	Identity Ref	Optional	ID of the ContingencyEpisode if held.

2.2 DocumentLine (Entity within FiscalDocument)

- **Purpose:** One billed good/service inside the document.
- **Lifecycle:** Frozen when the parent document is signed.

Attribute	Conceptual Type	Req/Opt	Description
LineIdentifier	Identifier	Required	Unique within the parent document.
ItemSnapshot	Value Object	Required	Frozen description, unit, and item details.
Quantity	Value Object	Required	Measured amount.
UnitPrice	Value Object	Required	Monetary amount per unit.
TaxLines	List of Values	Required	Computed tax categories and amounts for this line.
LineTotal	Value Object	Required	Computed gross line total.

2.3 ControlNumberSeries (Aggregate Root)

- **Purpose:** The single authority for gapless, ordered correlatives for one DTE Type at one issuing point.
- **Validation Rules:** Correlatives are strictly monotonic. Assignments are atomic. Never rewinds, even for rejected documents.

Attribute	Conceptual Type	Req/Opt	Description
SeriesIdentity	Unique Identifier	Required	Internal aggregate ID.
DTEType	String	Required	The DTE Type this series serves (e.g., '01').
IssuingPoint	Value Object	Required	Establishment and POS coordinates.

NextCorrelative	Integer	Required	The next gapless number to be handed out.
------------------------	---------	----------	---

2.4 Counterparty (Aggregate Root)

- **Purpose:** Reusable master data for potential document receptors.
- **Immutability Rules:** Highly mutable. However, edits here *never* cascade to documents already signed.

Attribute	Conceptual Type	Req/Opt	Description
CounterpartyIdentity	Unique Identifier	Required	Internal aggregate ID.
TaxIdentity	Value Object	Required	Type (consumer/taxpayer) and string value.
LegalName	String	Required	Commercial or legal name.
Address	Value Object	Required	Structured location (department, municipality, etc.).
IsActive	Boolean	Required	Soft-delete flag.

2.5 Item (Aggregate Root)

- **Purpose:** Reusable product/service master data.
- **Immutability Rules:** Edits here *never* cascade to already signed Document Lines.

Attribute	Conceptual Type	Req/Opt	Description
ItemIdentity	Unique Identifier	Required	Internal aggregate ID.
Description	String	Required	Human-readable product name.

DefaultUnitPrice	Value Object	Required	Non-negative baseline price.
TaxBehavior	Value Object	Required	Default tax category (e.g., standard IVA, exempt).
IsActive	Boolean	Required	Soft-delete flag.

3. Snapshot Strategy

The core of V1's fiscal compliance and data integrity relies on the **Snapshot Pattern**.

- **Mechanism:** When a Fiscal Document moves to Validated or Signed, it does not store a reference (ID) to the current Counterparty or Item. Instead, it copies the exact state of that data at that exact millisecond into embedded Value Objects (ReceptorSnapshot, IssuerSnapshot, ItemSnapshot).
- **Outcome:** If an administrative user changes a client's address or an item's price tomorrow, yesterday's documents remain perfectly intact and legally accurate. The snapshot is permanently immutable.

4. Specific Sub-Models

4.1 Evidence Data Model

- **Purpose:** A write-once, append-only vault of legal proof that outlives the transactional database.
- **Aggregate:** EvidenceBundle (1:1 with a Signed FiscalDocument).
- **Entities:** EvidenceArtifact.
 - *Payload Artifact:* The exact cryptographically signed JSON.
 - *Authority Response Artifact:* The raw success/rejection payload received from the authority.
 - *Stamp Artifact:* The extracted Acceptance Stamp.
 - *Visual Artifact:* The deterministic PDF/human-readable generation.
- **Integrity:** Every artifact requires a ContentHash (e.g., SHA-256) calculated at the moment of insertion. Artifacts can never be deleted, even if the document is invalidated.

4.2 Audit Data Model

- **Purpose:** Tamper-evident, chronological ledger of all significant business actions.
- **Aggregate:** AuditTrail (A global singleton or per-issuer construct).
- **Entities:** AuditEntry.
 - *Who:* Actor ID and Role.
 - *What:* Action performed (e.g., "Sign", "Submit", "Invalidate").

- *When*: Immutable timestamp.
- *On What*: Target identity (Generation Code, Configuration Key).
- *Chain*: CurrentHash and PreviousHash linking it strictly to the prior event to detect tampering.

4.3 Contingency Data Model

- **Purpose**: To manage lawful offline issuance during internet outages.
- **Aggregate**: ContingencyEpisode.
 - *State*: Open $\rightarrow$ Reconciling $\rightarrow$ Closed.
 - *Attributes*: Start/End Timestamps, Contingency Reason (authority unreachable), Notification Status.
- **Membership**: Documents signed during an Open episode are stamped with a ContingencyRef linking them to the episode. They are held in Contingency-Held status.
- **Reconciliation**: When connectivity restores, the episode dictates transmission order: Notification first, then the batch of held documents.

4.4 Control Number Management Model

- **Purpose**: To isolate the contention of monotonic sequence generation.
- **Structure**: A composite key maps to exactly one ControlNumberSeries. Key = (DTE Type + Establishment Code + POS Code).
- **Behavior**: The sequence is strictly a read-and-increment integer operation. Re-opening a validated document to draft discards the number forever to preserve the gapless audit rule.

4.5 Reference Data Model (Document Type Registry)

- **Purpose**: To remove hard-coded logic regarding what a "Factura" vs "CCF" is.
- **Aggregate**: DocumentTypeSpecification.
- **Attributes**: Type Code (01, 03), Bound Schema Version, Allowable Receptor Types, Mandated Fields.
- **Usage**: Read-only at runtime in V1. Issuance reads this model to dynamically validate documents.

5. Conceptual Entity Relationship Model

Note: All relationships are conceptual identity links, never live cascading foreign-key database constraints, to respect bounded contexts.

1. **FiscalDocument $\rightarrow$ DocumentTypeSpecification (N:1)**
 - Many documents reference exactly one Type Specification to govern their structural rules.
2. **FiscalDocument $\rightarrow$ DocumentLine (1:N)**
 - A document contains one or more lines. Lines cannot exist outside a document.
3. **FiscalDocument $\rightarrow$ EvidenceBundle (1:1)**
 - Exactly one bundle exists for every document that successfully reaches the Signed state.

4. **FiscalDocument $\rightarrow$ TransmissionRecord (1:N)**
 - A document has one Transmission Record for its initial issuance. (In V1, invalidations have their own independent record).
5. **ContingencyEpisode $\rightarrow$ FiscalDocument (1:N)**
 - An episode contains zero-to-many documents held during the outage. A document belongs to at most one episode.
6. **InvalidationEvent $\rightarrow$ FiscalDocument (0..1:1)**
 - A fiscal document may have a maximum of one accepted Invalidation Event targeting it. The event references the document via its Generation Code.
7. **User $\rightarrow$ AuditEntry (1:N)**
 - A user (Actor) generates multiple audit entries through their actions.
8. **FiscalDocument $\rightarrow$ Counterparty/Item (Historical/None)**
 - There is **no** structural relationship between a signed Fiscal Document and the Master Data tables. The document contains an embedded Value Object snapshot captured at issuance, completely severing the link to the live Counterparty or Item.